

NATURAL LANGUAGE PROCESSING FOR STRUCTURED TEXT GENERATION

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Objective:

- Generate song lyrics that bears semblance to a song by one of the artists from the dataset without memorizing lines

Mathematical Formulation Of LSTMs:

$$i_t = \sigma(W^i x_t + U^i h_{t-1})$$

$$f_t = \sigma(W^f x_t + U^f h_{t-1})$$

$$o_t = \sigma(W^o x_t + U^o h_{t-1})$$

$$\tilde{c}_t = \tanh(W^c x_t + U^c h_{t-1})$$

$$c_t = i_t \cdot \tilde{c}_t + f_t \cdot \tilde{c}_{t-1}$$

$$h_t = o_t \cdot \tanh(c_t)$$

Results

Generated text[1] :

keeps tellin '

as i admit it at night

send but

i knew onto but god knows that

i 're not a thing that had demons down

Generated text [2] :

Remember those walls I built

This is just another day

That he's wrong

You say he's not



Methods:

First

- Pre-trained model :3-layer AWD-LSTM model trained on 100 million tokens from Wikipedia articles

- Then trained on the dataset of song lyrics

- Freeze and train only on certain layers

- Fine-tuning

- **Drawback:** Still required larger amount of training data

- **Advantage:** Good vocabulary

Second

- Pre-trained model :1-layer-LSTM trained on a corpus of text written by Nietzsche

- Trained on the dataset of songs

- 40 epochs to ensure vocabulary retention

- **Drawback:** Model does not have as good a vocabulary as the first model

- **Advantage:** Learns grammar and sentence structure well

Future Scope:

- Improve the model such that it has good vocabulary and learns grammar and sentence structure well

- Generate lyrics that could be mistaken for lyrics from songs by popular artists

- Generate a full, well-structured, meaningful song

Conclusion:

- AI can generate lyrics in the style of an artist the model has been trained on

- In future AI can replace song writers and write lyrics which will become super hits by learning the most popular songs

